

Mathematics of Neural Networks

Linear Algebra: Revisiting Linear Regression (STS 311)

You have seen Linear Regression in terms of scalars. But we will rederive linear regression in terms of vectors and Matrices.

Notation :

Scalar

Handwritten a, b, c
 α, β, γ

Print

a, b, c
 α, β, γ

Vectors

a, b, c

a, b, c

↑ Bolded
small letters

Matrices

A, B, C

A, B, C

Geometric vector

$\vec{a}, \vec{b}, \vec{c}$

$$\vec{a} = 1\hat{i} + 2\hat{j} + 3\hat{k}$$

$$= 1 \underbrace{\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}}_{\hat{i}} + 2 \underbrace{\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}}_{\hat{j}} + 3 \underbrace{\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}_{\hat{k}}$$

$$= \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Linear algebra
vector

$$\underline{a} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \leftarrow \begin{matrix} \text{column} \\ \text{matrix} \end{matrix}$$

3x1

Dot product

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

$$\vec{a} = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

$$\vec{b} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$$\checkmark \quad \underline{a}^T \underline{b} = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}_{1 \times 3} \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}_{3 \times 1} \\ = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Transpose of a product

$$(AB)^T = B^T A^T$$

$$(ABC)^T = C^T B^T A^T$$

$$(\underline{a}^T B \underline{c})^T = \underline{c}^T B^T \underline{a}$$

$$(A^T)^T = A$$

Inverse

$$(AB)^{-1} = B^{-1} A^{-1}$$

Common Mistake : Using division with Matrices

$$A\underline{x} = \underline{b} \Rightarrow \boxed{\underline{x} = \frac{\underline{b}}{A}} \times$$

$$A \underline{x} = \underline{b}$$

Left multiply both side with A^{-1}

$$A^{-1} A \underline{x} = A^{-1} \underline{b}$$

$$\Rightarrow \underline{x} = A^{-1} \underline{b}$$

~~$$\underline{x} = \underline{b} A^{-1}$$~~

($\underline{y}$ it exists)

$$\underline{x}^T A = \underline{b}^T$$

Right multiply

$$\underline{x}^T = \underline{b}^T A^{-1}$$

~~$$\underline{x}^T = A^{-1} \underline{b}^T$$~~

$\underline{a}, \underline{b}, \underline{c} \equiv$ column vectors $\begin{bmatrix} \end{bmatrix}_{n \times 1}$

$\underbrace{\begin{bmatrix} \text{row vector} \end{bmatrix}}_{\underline{a}^T, \underline{b}^T, \underline{c}^T}_{1 \times n}$

$$\begin{array}{|c|} \hline \underline{a} B \quad \times \\ \hline \underline{a} \quad \underline{b}^T \quad \checkmark \\ \hline \underline{a}^T B \quad \checkmark \\ \hline \end{array}$$

$$\begin{bmatrix} \end{bmatrix}_{n \times 1} \begin{bmatrix} \end{bmatrix}_{1 \times m}$$

$$\begin{array}{|c|} \hline \underline{B} \underline{c} \\ \hline \underline{B} \underline{c}^T \\ \hline \underline{b} \underline{c}^T \\ \hline \end{array}$$

✓ ✗ ✓

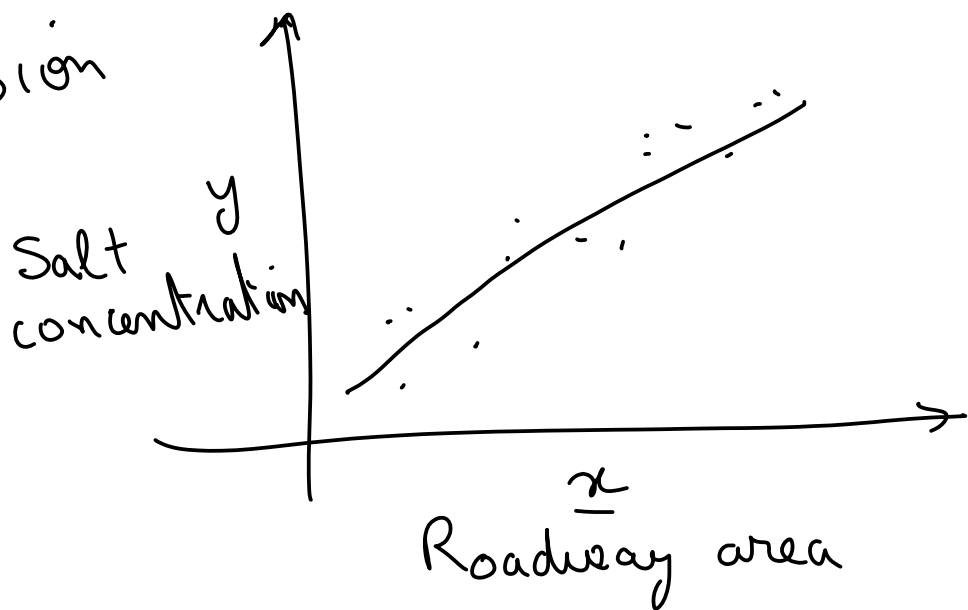
$$\underbrace{\underline{a}^T \underline{X} \underline{B} \underline{y}}_{1 \times n \quad n \times m \quad m \times 1} = 1 \times 1$$

scalar

Linear Regression

Dataset

$$\mathcal{D} = \{ (\underline{x}_1, y_1), (\underline{x}_2, y_2), \dots, (\underline{x}_n, y_n) \}$$



Model

$$\hat{y} = \underbrace{\underline{w}^T \underline{x}}_{f(\underline{x})} + w_0$$

(Red arrows point from $\underline{w}^T$ and w_0 to the word 'Parameters' in the adjacent block)

Parameters

$$\hat{y} = mx + c$$

Loss

$$L(\mathcal{D}; w, w_0) = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Least squares loss

Training

minimizing w, w_0

$$L(\mathcal{D}; w, w_0)$$

1. Write everything in terms of vectors (Vectorization)

$$\underline{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \quad \rightarrow \quad ||\underline{v}|| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

$$\underline{v}^T \underline{v} = \begin{bmatrix} v_1 & v_2 & \dots & v_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$= v_1^2 + v_2^2 + \dots + v_n^2$$

$$\underline{v}^T \underline{v} = ||\underline{v}||^2$$

$$L(\underline{D}, \underline{w}, w_s) = \sum_{i=1}^n \underbrace{(y_i - \hat{y}_i)^2}_{e_i = y_i - \hat{y}_i}$$

$$= \sum_{i=1}^n e_i^2 = ||\underline{e}||^2$$

$$\underline{e} = \begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}_{n \times 1}$$

want to write in terms of $\underline{y}$ and some X

$$\underline{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Residual vector

$$\underline{e} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\underline{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}_{n \times 1}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \underline{w}^T \underline{x}_1 + w_0 \\ \underline{w}^T \underline{x}_2 + w_0 \\ \vdots \\ \underline{w}^T \underline{x}_n + w_0 \end{bmatrix} = \begin{bmatrix} \underline{w}^T \underline{x}_1 \\ \underline{w}^T \underline{x}_2 \\ \vdots \\ \underline{w}^T \underline{x}_n \end{bmatrix}_{n \times 1} + \begin{bmatrix} w_0 \\ \vdots \\ w_0 \end{bmatrix}_{n \times 1}$$

$$[\underline{w}^T \quad \underline{w}^T \quad \underline{w}^T]$$

$$\begin{bmatrix} \underline{w}^T \end{bmatrix}_{1 \times nd} \begin{bmatrix} \underline{x}_1 \\ \underline{x}_2 \\ \vdots \\ \underline{x}_n \end{bmatrix}_{nd \times 1}$$

$$\underline{w}_{1 \times d}^T \begin{bmatrix} \underline{x}_1 \\ \underline{x}_2 \\ \vdots \\ \underline{x}_n \end{bmatrix}_{nd \times 1}$$

$$\boxed{\underline{a}^T \underline{b} = \underline{b}^T \underline{a}}$$

$$\underline{x} \in \mathbb{R}^d$$

$$\underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}_{d \times 1}$$

$$\underline{w} \in \mathbb{R}^d$$

$$\underline{w}_{1 \times d}^T \underline{x}_{d \times 1}$$

$$\begin{bmatrix} \underline{w}^T \underline{x}_1 \\ \underline{w}^T \underline{x}_2 \\ \vdots \\ \underline{w}^T \underline{x}_n \end{bmatrix}_{n \times 1} = \begin{bmatrix} \underline{x}_1^T \underline{w} \\ \underline{x}_2^T \underline{w} \\ \vdots \\ \underline{x}_n^T \underline{w} \end{bmatrix}_{n \times 1} = \begin{bmatrix} \underline{x}_1^T_{1 \times d} \\ \underline{x}_2^T_{1 \times d} \\ \vdots \\ \underline{x}_n^T_{1 \times d} \end{bmatrix}_{n \times d} \underline{w}_{d \times 1}$$

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \underline{x}_1^T \\ \underline{x}_2^T \\ \vdots \\ \underline{x}_n^T \end{bmatrix}_{n \times d} \underline{w}_{d \times 1} + \begin{bmatrix} 1 \cdot w_0 \\ 1 \cdot w_0 \\ \vdots \\ 1 \cdot w_0 \end{bmatrix}_{n \times 1}$$

Want to absorb both the parameters w and w_0 into the same vector

$$= \begin{bmatrix} \underline{x}_1^T & 1 \\ \underline{x}_2^T & 1 \\ \vdots & \vdots \\ \underline{x}_n^T & 1 \end{bmatrix}_{n \times (d+1)} \begin{bmatrix} \underline{w} \\ w_0 \end{bmatrix}_{(d+1) \times 1}$$

X $\tilde{w}_{(d+1) \times 1}$

Affine
 $\hat{y}_i = \underline{w}^T \underline{x}_i + w_0$
 $= \underbrace{[\underline{w}^T \quad w_0]}_{1 \times (d+1)} \underbrace{\begin{bmatrix} \underline{x}_i^T \\ 1 \end{bmatrix}}_{\substack{\text{Linear} \\ \text{Homogeneous} \\ \text{vector}}}$
 Homogenization trick

$= X \tilde{w}$

$$\underline{e} = \underline{y} - X \tilde{w}$$

$\downarrow$ Dataset $\uparrow$ Parameters

$$\underline{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad X = \begin{bmatrix} \underline{x}_1^T & 1 \\ \underline{x}_2^T & 1 \\ \vdots & \vdots \\ \underline{x}_n^T & 1 \end{bmatrix}, \quad \tilde{w} = \begin{bmatrix} \underline{w} \\ w_0 \end{bmatrix}$$

$$\text{minimize } L(D; \underline{w}, w_0)$$

$$\text{minimize } \underline{e}^T \underline{e}$$

$$\text{minimize}_{\underline{\tilde{w}}} \frac{(\underline{y} - X \underline{\tilde{w}})^T (\underline{y} - X \underline{\tilde{w}})}{\downarrow}$$

$$\|\underline{\tilde{w}}\|^2 + \underline{y}^T \underline{\tilde{w}}^T X^T - \underline{y}^T \underline{\tilde{w}} X + \|\underline{y}\|^2$$

scalar
 $\underline{e}^T \underline{e}$

$$= (\underline{y}^T - (X \underline{\tilde{w}})^T) (\underline{y} - X \underline{\tilde{w}})$$

$$= (\underline{y}^T - \underline{\tilde{w}}^T X^T) (\underline{y} - X \underline{\tilde{w}})$$

$$= \underbrace{\underline{y}^T \underline{y}}_{\text{scalar}} - \underbrace{\underline{\tilde{w}}^T X^T \underline{y}}_{\text{scalar}} - \underbrace{\underline{y}^T X \underline{\tilde{w}}}_{\text{scalar}} + \underbrace{\underline{\tilde{w}}^T X^T X \underline{\tilde{w}}}_{\text{matrix}}$$

scalars 1×1 matrices
 are symmetric.

$$(\underline{\tilde{w}}^T X^T \underline{y})^T = \underline{y}^T X \underline{\tilde{w}}$$

$$(A+B)^T = A^T + B^T$$

$$(AB)^T = B^T A^T$$

$$AB \neq BA$$

$$[a]^T = [a]$$

$$\underline{e}^T \underline{e} = \underline{y}^T \underline{y} - 2 \underline{y}^T X \underline{\tilde{w}} + \underline{\tilde{w}}^T X^T X \underline{\tilde{w}}$$

Quadratic function

Scalar version

$$\underline{e}^T \underline{e} = \underline{y}^T \underline{y} - 2\alpha w + \kappa^2 w^2$$

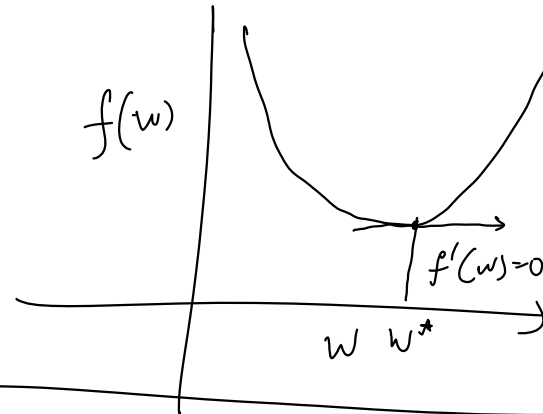
Quadratic function

How do you minimize a Quad fund?

$$\frac{d}{dw} (\quad \quad \quad) = 0$$

$$-2\alpha + 2\kappa^2 w^* = 0$$

$$w^* = \frac{\alpha}{\kappa^2}$$



Vector calculus $\equiv$ Multi variate calculus

↖
reshape into vectors

$$f(x, y) = f\left(\begin{bmatrix} x \\ y \end{bmatrix}\right)$$

multivariate

vector function

Two conventions

① Gradient convention

② Jacobian convention

$$f(\underline{x}) : \mathbb{R}^d \mapsto \mathbb{R}$$

scalar valued
vector function

$$\frac{\partial}{\partial \underline{x}} f(\underline{x}) = \begin{bmatrix} \frac{\partial}{\partial x_1} f(\underline{x}) \\ \vdots \\ \frac{\partial}{\partial x_d} f(\underline{x}) \end{bmatrix}_{d \times 1} = \nabla_{\underline{x}} f(\underline{x}) \quad \text{Gradient convention}$$

$$\frac{\partial f(\underline{x})}{\partial x_1} \quad \frac{\partial f(\underline{x})}{\partial x_2} \quad \dots \quad \frac{\partial f(\underline{x})}{\partial x_d} \quad \underline{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_d \end{bmatrix}$$

$$\frac{\partial}{\partial \underline{x}} f(\underline{x}) = \left[\frac{\partial f(\underline{x})}{\partial x_1}, \dots, \frac{\partial f(\underline{x})}{\partial x_d} \right]_{1 \times d} \quad \text{Jacobian convention}$$

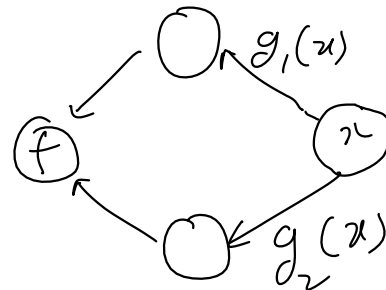
Multi variate calculus chain rule

$$\begin{aligned} \frac{\partial}{\partial x} f(g_1(x), g_2(x)) \\ = \frac{\partial f}{\partial g_1} \cdot \frac{\partial g_1}{\partial x} + \frac{\partial f}{\partial g_2} \cdot \frac{\partial g_2}{\partial x} \end{aligned}$$

$$= \underbrace{\begin{bmatrix} \frac{\partial f}{\partial g_1} & \frac{\partial f}{\partial g_2} \end{bmatrix}}_{\frac{\partial f}{\partial \underline{g}}} \underbrace{\begin{bmatrix} \frac{\partial g_1}{\partial x} \\ \frac{\partial g_2}{\partial x} \end{bmatrix}}_{\frac{\partial \underline{g}(x)}{\partial x}}$$

$$\boxed{\frac{\partial}{\partial x} f(\underline{g}(x)) = \left[\frac{\partial f}{\partial \underline{g}} \right] \left[\frac{\partial \underline{g}(x)}{\partial x} \right]}$$

Chain rule for Jacobian convention



$$\underline{g}(x) = \begin{bmatrix} g_1(x) \\ g_2(x) \end{bmatrix}$$

↑
vector valued
scalar function

$$\begin{matrix} f(\underline{g}(x)) \\ \uparrow \quad \uparrow \end{matrix}$$

Jacobian
(matrix) of a vector-valued vector function

$$\underline{f}(\underline{x}) : \mathbb{R}^d \rightarrow \mathbb{R}^m$$

$$J_{\underline{x}} \underline{f}(\underline{x}) = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \dots & \frac{\partial f_1}{\partial x_d} \\ \frac{\partial f_2}{\partial x_1} & & & \\ \vdots & & & \\ \frac{\partial f_m}{\partial x_1} & \dots & \frac{\partial f_m}{\partial x_d} \end{pmatrix}_{m \times d}$$

Chain rule

$$J_{\underline{x}} \underline{f}(\underline{g}(\underline{x})) = \frac{\partial}{\partial \underline{x}} \underline{f}(\underline{g}(\underline{x})) = \frac{\partial \underline{f}}{\partial \underline{g}} \frac{\partial \underline{g}}{\partial \underline{x}} = \left[J_{\underline{g}} \underline{f}(\underline{g}) \right] \left[J_{\underline{x}} \underline{g}(\underline{x}) \right]$$